DATA SCIENCE &
ARTIFICIAL INTELLIGENCE

Probabilistic Machine Learning

Variational Inference

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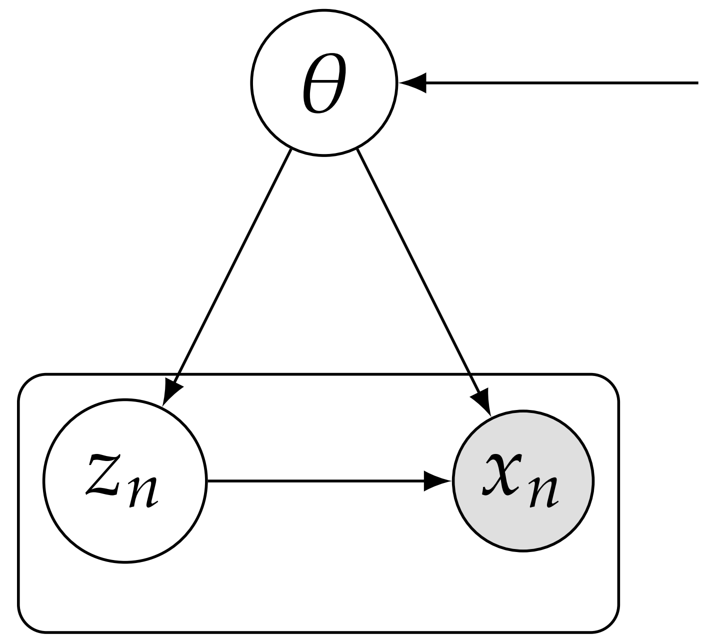
The VI formulation



$p(x, z)$

local latent variables z_i

global latent variables θ

goal: compute $p(z \,|\, \underline{x})$ and $p(\underline{x}) = \int p(\underline{x}, z) dz.$

$$\log p(\underline{x}) = \mathcal{L}(q) + KL[q(z)||p(z|\underline{x})]$$

$$\mathcal{L}(q) = \int q(z) \left[\log p(\underline{x}, z) - \log q(z)\right] dz = \mathbb{E}_q[\log p(\underline{x}, z) - \log q(z)]$$

$$KL[q(z)||p(z|\underline{x})] = -\int q(z) \left[\log p(z|\underline{x}) - \log q(z)\right] dz$$

The VI formulation

- the q that minimizes the KL-divergence is the same as the one maximizing the ELBO $\mathscr{L}(q)$
- $\mathscr{L}(q)$ is maximum when $KL[q||p] = 0$, i.e. $q = p(z|\underline{x})$

However, in most of the cases the computation of $p(z|\underline{x})$ is intractable.

restrict $q(z)$ to tractable family of distributions Q

e.g. parametric with parameters λ: $Q = \{q(z\,|\,\lambda), \lambda \in \mathbb{R}^k\}$

$\mathscr{L}(q) = \mathscr{L}(\lambda)$ is typically non convex

Mean Field VI

we assume that z can be decomposed in M different blocks of variables $z = (z_1, \ldots, z_M)$

$$q(z) = \prod_{i=1}^{M} q_i(z_i)$$

$q_i(z_i) = q_i$ for brevity.

$$\mathcal{L}(q) = \mathbb{E}\left[\log p(\underline{x}, z) - \log q(z)\right] =$$

$$\int \prod_i q_i [\log p(\underline{x}, z) - \sum_i \log q_i] dz =$$

$$= \int q_j \left[\int \log p(\underline{x}, z) \prod_{i \neq j} q_i dz_i \right] dz_j - \int q_j \log q_j dz_j + \text{const}$$

$$= \int q_j \mathbb{E}_{i \neq j}[\log p(\underline{x}, z)] dz_j - \int q_j \log q_j dz_j + \text{const}$$

Mean Field VI

define $\log \tilde{p}(\underline{x}, z_j) = \mathbb{E}_{i \neq j}[\log p(\underline{x}, z)] + \text{const}$

$$\mathscr{L}(q_j(z_j)) = \mathbb{E}_{q_j}[\log \tilde{p}(\underline{x}, z_j) - \log q_j] + \text{ const},$$

$$\text{hence } \mathscr{L}(q_j(z_j)) = -KL[q_j || \tilde{p}(\underline{x}, z_j)] + \text{ const}$$

hence $q_j^\star(z_j) = \tilde{p}(\underline{x}, z_j)$

$$q_j^\star(z_j) = \frac{\exp(\mathbb{E}_{i \neq j}[\log p(\underline{x}, z)])}{\int \exp\left(\mathbb{E}_{i \neq j}[\log p(\underline{x}, z)]\right) dz_j}$$

We have M lower bounds $\mathscr{L}(q_j)$

If we can compute analytically the expectation $\mathbb{E}_{i \neq j}[\log p(\underline{x}, z)]$

We can optimize the ELBO by coordinate ascent on q_j

Example: Factorization of Gaussians

$p(z) = \mathcal{N}(z \,|\, \mu, \Lambda^{-1})$ $\qquad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$ $\qquad \Lambda$ precision

$$q(z) = q(z_1)q(z_2)$$

$$\log q_1^\star(z_1) = \mathbb{E}_{z_2}[\log p(z)] + \text{ const}$$

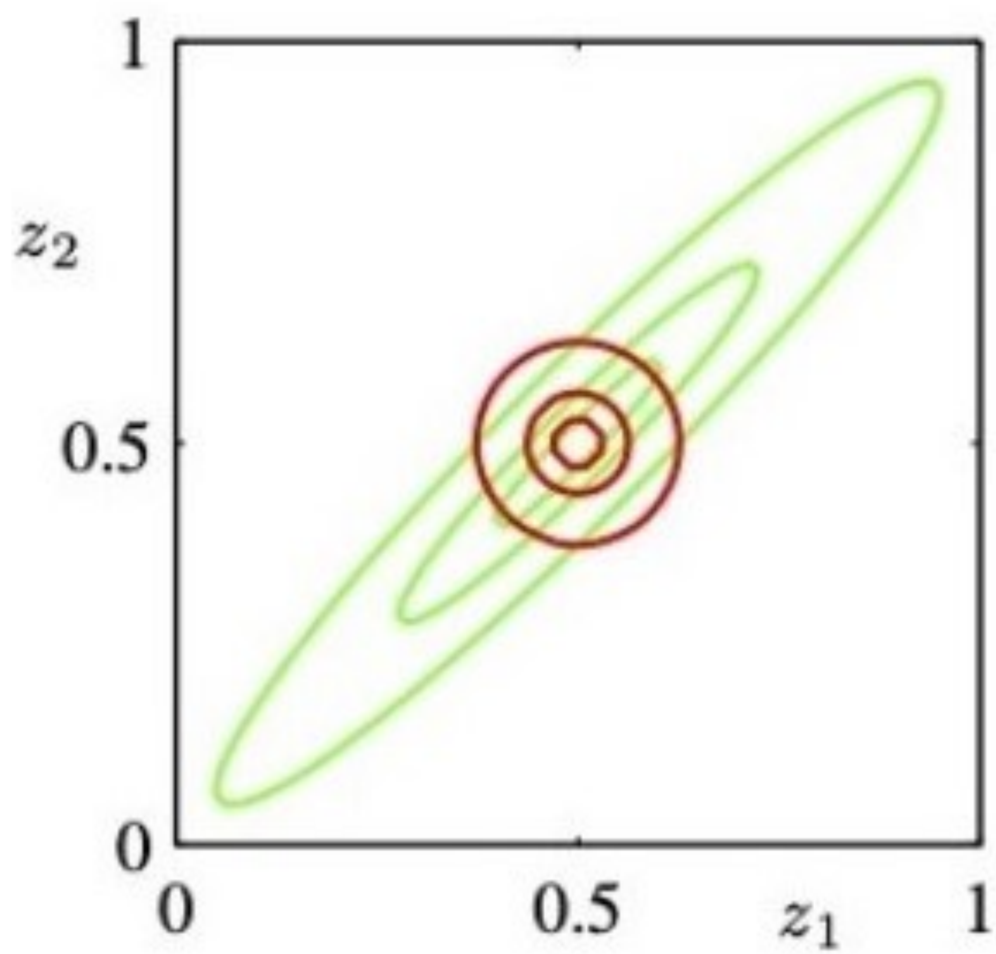
$$= \mathbb{E}_{z_2}[-\frac{1}{2}(z_1 - \mu_1)^2\Lambda_{11} - (z_1 - \mu_1)\Lambda_{12}(z_2 - \mu_2)] + \text{ const}$$

$$= -\frac{1}{2}(z_1 - \mu_1)^2\Lambda_{11} - (z_1 - \mu_1)\Lambda_{12}(\mathbb{E}[z_2] - \mu_2) + \text{ const}$$

Example: Factorization of Gaussians

$q_1^\star(z_1)$ is Gaussian $\mathcal{N}(z_1 \mid m_1, \Lambda_{11}^{-1})$: $m_1 = \mu_1 - \Lambda_{11}^{-1}\Lambda_{12}(\mathbb{E}[z_2] - \mu_2)$

$$q_2^\star(z_2) = \mathcal{N}(z_2|m_2, \Lambda_{22}^{-1}), \qquad m_2 = \mu_2 - \Lambda_{22}^{-1}\Lambda_{21}(\mathbb{E}[z_1] - \mu_1)$$



Direct and Inverse KL

Direct KL: minimise w.r.t. q the functional $KL[q\,||\,p]$

Inverse KL: minimise w.r.t. q the functional $KL[p\,||\,q]$

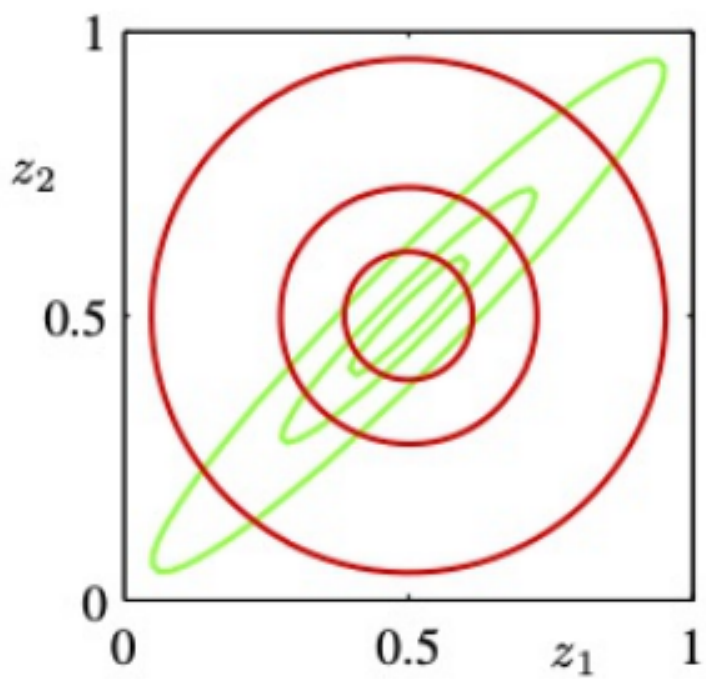


Figure 2: Inverse KL divergence $KL[p||q]$ is zero avoiding.

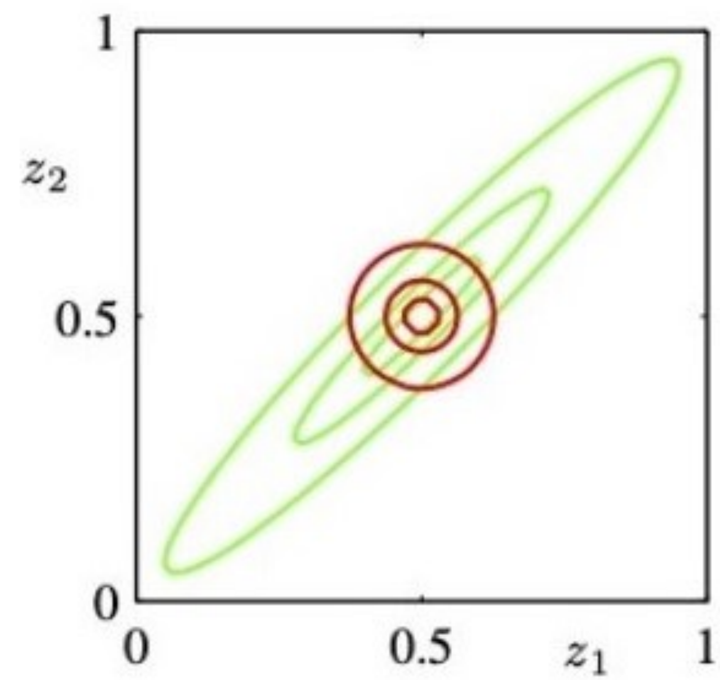
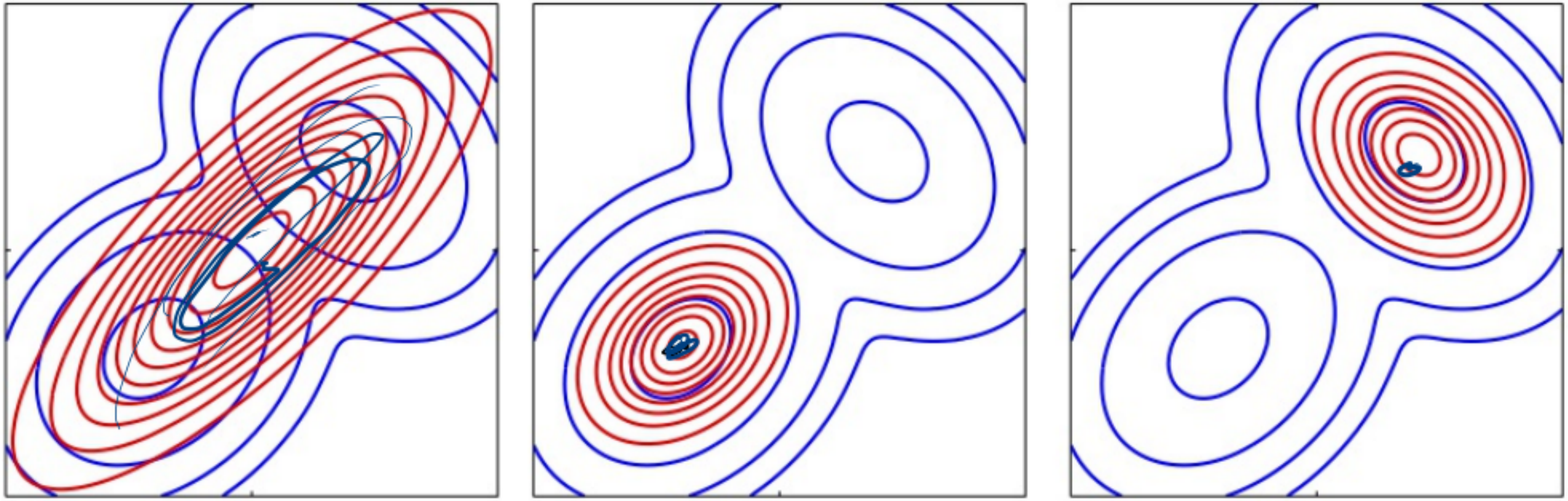


Figure 3: Direct KL divergence $KL[q||p]$ is zero forcing.

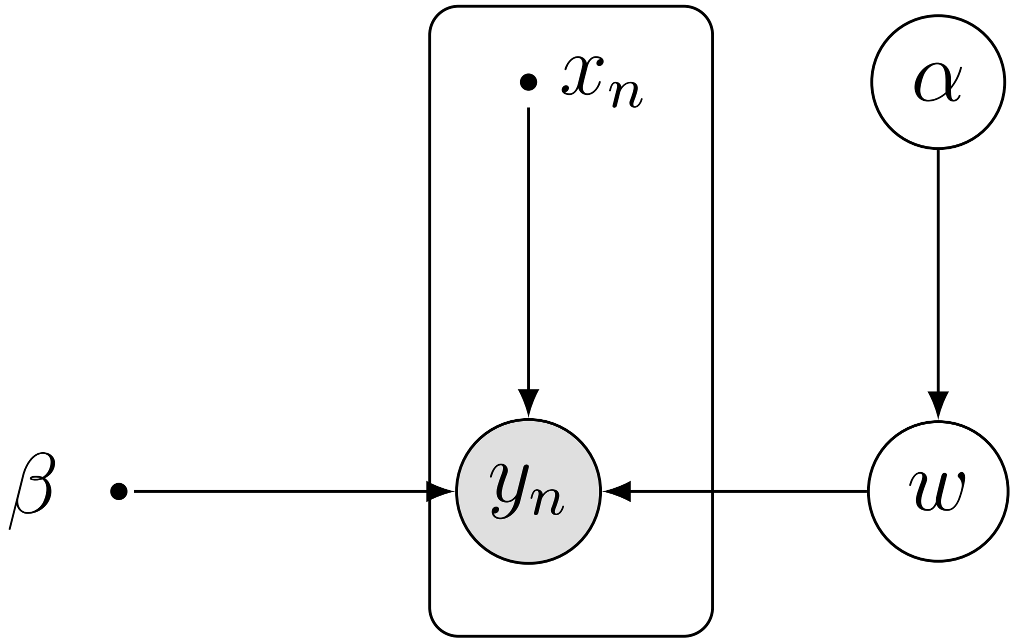
direct KL is zero-forcing: $p(z) \approx 0$ implies $q(z) \approx 0$

inverse KL is zero-avoiding: $q(z) > 0$ implies $p(z) > 0$

Direct and Inverse KL



Variational Linear Regression



hyper prior on α

$$p(y, w, \alpha) = p(y|w)p(w|\alpha)p(\alpha)$$

$$p(y|w) = \prod_{n=1}^{N} \mathcal{N}(y_n|w^T\phi(x_n), \beta^{-1})$$

$$p(w|\alpha) = \mathcal{N}(w|0, \alpha^{-1}I)$$

$$p(\alpha) = \mathrm{Gamma}(\alpha|a_0, b_0) = \frac{1}{\Gamma(a_0)} b_0^{a_0} \alpha^{a_0-1} e^{-b_0\alpha}$$

Variational Linear Regression

$$p(w, \alpha | y) \qquad\qquad q(w, \alpha) = q(w)q(\alpha)$$

$$\begin{aligned} q^\star(\alpha) &= \mathbb{E}_w[\log p(y, w, \alpha)] + \text{ const} \\ &= \log p(\alpha) + \mathbb{E}_w[\log p(w|\alpha)] + \text{ const} \\ &= (a_0 - 1)\log\alpha - b_0\alpha + \frac{M}{2}\log\alpha - \frac{\alpha}{2}\mathbb{E}[w^T w] + \text{ const} \end{aligned}$$

$$q^\star(\alpha) = \text{Gamma}(\alpha | a_N, b_N)$$

$$a_N = a_0 + \frac{M}{2}$$

$$b_N = b_0 + \frac{1}{2}\mathbb{E}_q[w^T w]$$

Variational Linear Regression

$$\log q^\star(w) = \log p(y|w) + \mathbb{E}_\alpha[\log p(w|\alpha)] + \text{ const}$$

$$= -\frac{\beta}{2}\sum_{n=1}^{N}[w^T\phi(x_n) - y_n]^2 - \frac{1}{2}\mathbb{E}_\alpha w^T w + \text{ const}$$

$$q^\star(w) = \mathcal{N}(w|m_N, S_N)$$

$$m_N = \beta S_N \Phi^T y$$

$$S_N = (\mathbb{E}[\alpha]I + \beta\Phi^T\Phi)^{-1}$$

$$\mathbb{E}[\alpha] = \frac{a_N}{b_N}$$

$$\mathbb{E}[w^T w] = m_N^T m_N + \text{Trace}(S_N)$$

Black Box Variational Inference

Parametric family $q(z \mid \lambda) \in Q$

+

Monte Carlo estimate of the gradient of the ELBO

+

Stochastic Gradient Ascent

$$\mathscr{L}(\lambda) = \mathbb{E}_{q(z|\lambda)}[\log p(x, z) - \log q(z|\lambda)]$$

How to compute $\nabla \mathscr{L}(\lambda)$?

Reparametrization Trick

Idea: express $z = g_{\nu}(\varepsilon)$ with $\varepsilon \sim \hat{q}(\varepsilon)$ independent of λ

$$\hat{\lambda} \stackrel{\smile}{=} (\lambda, \nu)$$

$$\mathscr{L}(\hat{\lambda}) = \mathbb{E}_{\hat{q}(\varepsilon)}[\log p(x, g_{\nu}(\varepsilon)) - \log q(g_{\nu}(\varepsilon)|\lambda)]$$

$$\nabla_{\hat{\lambda}}\mathscr{L}(\hat{\lambda}) = \mathbb{E}_{\hat{q}(\varepsilon)}[\nabla_{\hat{\lambda}} \log p(x, g_{\nu}(\varepsilon)) - \nabla_{\hat{\lambda}} q(g_{\nu}(\varepsilon)|\lambda)]$$

$$\nabla_{\hat{\lambda}}\mathscr{L}(\hat{\lambda}) = \frac{1}{S}\sum_{s=1}^{S} G(\varepsilon_s) \qquad\qquad \varepsilon_j \sim \hat{q}(\varepsilon)$$

Typically: $q(z \,|\, \lambda) = \mathcal{N}(z \,|\, \mu, \sigma^2)$

Hence $z = g_{\nu}(\varepsilon) = \mu + \sigma \cdot \varepsilon$, with $\varepsilon \sim \hat{q}(\varepsilon) = \mathcal{N}(\varepsilon \,|\, 0,1)$

Non-Reparameterizable q(z)

$$\begin{aligned}\nabla_\lambda \mathcal{L}(\lambda) &= \nabla_\lambda \mathbb{E}_{q(z|\lambda)}[\log p(x,z) - \log q(z|\lambda)] \\ &= \nabla_\lambda \int q(z|\lambda)[\log p(x,z) - \log q(z|\lambda)]dz\end{aligned}$$

$$\begin{aligned}\nabla_\lambda \mathcal{L}(\lambda) \quad &= \quad \int \nabla_\lambda [\log p(x,z) - \log q(z|\lambda)] q(z|\lambda) dz \\ &+ \quad \int \nabla_\lambda q(z|\lambda)[\log p(x,z) - \log q(z|\lambda)]dz\end{aligned}$$

Non-Reparameterizable q(z)

First term

$$\int \nabla_\lambda [\log p(x,z) - \log q(z|\lambda)] q(z|\lambda) dz$$

$$\nabla_\lambda [\log p(x,z)] = 0$$

$$\begin{aligned}
\int \nabla_\lambda [\log p(x,z) - \log q(z|\lambda)] q(z|\lambda) dz &= -\mathbb{E}_q [\nabla_\lambda \log q(z|\lambda)] \\
&= -\mathbb{E}_q \left[\frac{\nabla_\lambda q(z|\lambda)}{q(z|\lambda)} \right] = \int \frac{\nabla_\lambda q(z|\lambda)}{q(z|\lambda)} q(z|\lambda) dz \\
&= \int \nabla_\lambda q(z|\lambda) dz \\
&= \nabla_\lambda \int q(z|\lambda) dz = \nabla_\lambda 1 = 0
\end{aligned}$$

Non-Reparameterizable q(z)

second term $$\int \nabla_\lambda q(z|\lambda)[\log p(x,z) - \log q(z|\lambda)]dz$$

$$\nabla_\lambda[\log q(z|\lambda)] = \frac{\nabla_\lambda q(z|\lambda)}{q(z|\lambda)} \qquad \nabla_\lambda q(z|\lambda) = \nabla_\lambda[\log q(z|\lambda)]q(z|\lambda)$$

$$\begin{aligned}\nabla_\lambda \mathcal{L}(\lambda) &= \int q(z|\lambda)\nabla_\lambda \log q(z|\lambda)[\log p(x,z) - \log q(z|\lambda)]dz \\ &= \mathbb{E}_q[\nabla_\lambda \log q(z|\lambda)[\log p(x,z) - \log q(z|\lambda)]]\end{aligned}$$

$$z_s \sim q(z|\lambda) \qquad \nabla_\lambda \mathcal{L}(\lambda) \approx \frac{1}{S}\sum_{s=1}^{S} \nabla_\lambda \log q(z_s|\lambda)[\log p(x,z_s) - \log q(z_s|\lambda)]$$

issue: high variance

Rao-Blackwellization

Goal: evaluate $\mathbb{E}_{xy}[J(x,y)]$ of $J(x,y)$

$\hat{J}(x) = \mathbb{E}_y[J(x,y) \mid x]$ hence $\mathbb{E}_x[\hat{J}(x)] = \mathbb{E}_{xy}[J(x,y)]$

$$Var[\hat{J}(x)] = Var[J(x,y)] - \mathbb{E}[(J(x,y) - \hat{J}(x))^2] < Var[J(x,y)]$$

$$q(z \mid \lambda) = \prod_{i=1}^{N} q(z_i \mid \lambda_i)$$ consider gradient w.r.t. λ_i

- $q_{(i)}$: marginal of $q(z|\lambda)$ on the terms that form the Markov Blanket $z_{(i)}$ in p
- $p_i(x, z_{(i)})$ the product of factors of $p(x|z)$ depending on $z_{(i)}$

$$\hat{\nabla}_{\lambda_i}[\mathcal{L}] := \mathbb{E}_{q(i)}[\nabla_{\lambda_i}\mathcal{L}(z_i)] = \mathbb{E}_{q(i)}\left[\nabla_{\lambda_i}\log q(z_i \mid \lambda_i)[\log p_i(x, z_{(i)}) - \log q(z_i \mid \lambda_i)]\right]$$

Control Variates

Goal: compute $\mathbb{E}_q[f]$

Idea: define $\hat{f}$ such that $\mathbb{E}_q[\hat{f}] = \mathbb{E}_q[f]$ and $VAR_q[\hat{f}] < VAR_q[f]$

How? choose h such that $\mathbb{E}[h] < \infty$, then

$$\hat{f}_a(z) = f(z) - a(h(z) - \mathbb{E}[h(z)])$$

then $\mathbb{E}_q[\hat{f}] = \mathbb{E}[f]$ and $Var[\hat{f}] = Var[f] - 2aCov[f, h] + a^2 Var[h]$

fix a to $a^\star$ that minimises the variance: $a^\star = \dfrac{COV(f, h)}{VAR(h)}$

Control Variates

Our scenario: from Rao-Blackwellization

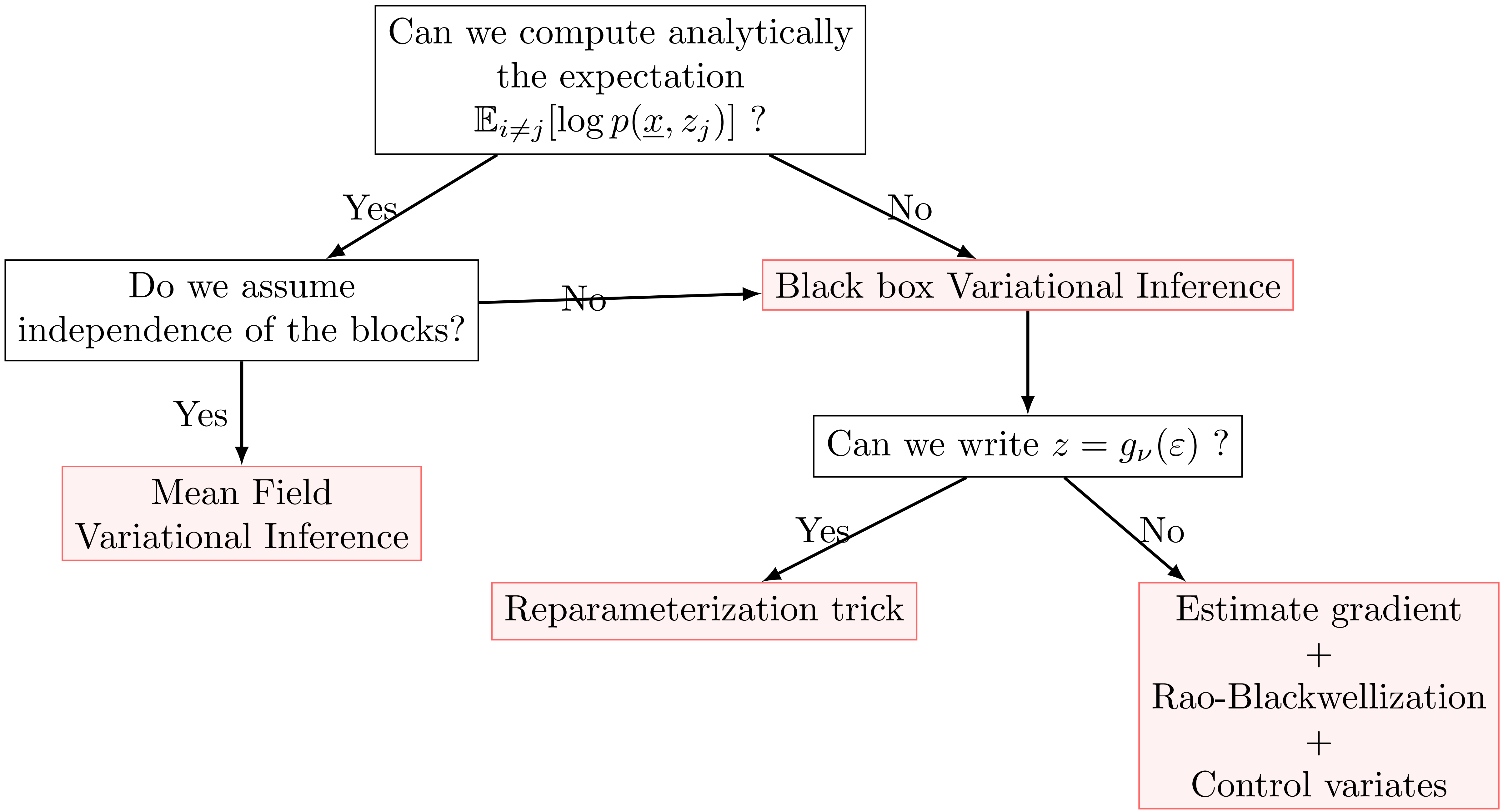
$$f_i(z) = \nabla_{\lambda_i} \log q(z_i \,|\, \lambda_i)[\log p_i(x, z_{(i)}) - \log(z_i \,|\, \lambda_i)]$$

choose $h_i(z) = \nabla_{\lambda_i} \log q(z_i \,|\, \lambda_i)$ such that $\mathbb{E}[h_i(z)] = 0$

sample $z_s \sim q_{(i)}(z \,|\, \lambda)$, estimate $\hat{a}_i^\star$ from the samples, then

$$\hat{\nabla}_{\lambda_i} = \frac{1}{S} \sum_{s=1}^{S} \nabla_{\lambda_i} \log q(z_i \,|\, \lambda_i)[\log p_i(x, z_s) - \log q_i(z_s \,|\, \lambda_i) - \hat{a}_i^\star]$$

VI - recap


Can we compute analytically the expectation $\mathbb{E}_{i \neq j}[\log p(\underline{x}, z_j)]$?
Yes
No
Do we assume independence of the blocks?
No
Black box Variational Inference
Yes
Mean Field Variational Inference
Can we write $z = g_\nu(\varepsilon)$?
Yes
No
Reparameterization trick
Estimate gradient + Rao-Blackwellization + Control variates